

# Northstar Analytics Dashboard

Product Requirements Document

|                |                                                |
|----------------|------------------------------------------------|
| Document ID    | PRD-NS-2026                                    |
| Version        | 3.1 (final pre-freeze)                         |
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| Eng Lead       | Megan Shaw, megan.shaw@gmail.com               |
| Designer       | Tomoko Hayashi                                 |
| Data Science   | Ryan Iyer                                      |
| Feature freeze | October 9, 2026                                |
| Ship target    | November 13, 2026                              |
| Status         | In beta with three pilot customers             |

## 1. Executive Summary

Northstar is Vantage Loop's first enterprise-grade analytics dashboard, designed to give Growth and Analytics pod customers self-serve visibility into product engagement, retention, and funnel performance. The dashboard aggregates event data from the Vantage Loop platform and surfaces it through configurable views, cohort tables, and export tools tuned for the workflows of enterprise data teams. Northstar addresses the leading churn signal from the 2025 renewal cohort: customers who could not answer their own retention questions without filing a support ticket.

The product is currently in beta with three pilot accounts (ENT-PC-001, ENT-PC-002, ENT-PC-003) and is on track for general availability on November 13, 2026, following a feature freeze on October 9, 2026. This document is the authoritative spec for all in-scope features. Post-launch items are documented in the backlog (see X12.xlsx).

## 2. Problem Statement

Enterprise customers at Vantage Loop manage product portfolios with multiple teams, user cohorts, and feature rollout schedules. Today, answering a question such as how many enterprise-tier users activated the Projects feature in the last 30 days requires either a data team query or a support ticket, both of which carry a 2 to 3 business-day turnaround. This friction concentrates analytics knowledge in the hands of Vantage Loop engineers and creates a dependency that scales poorly as the customer base grows. The result is low data confidence at the customer level, longer time to insight for product decisions, and measurable churn risk among the analytics-forward the event pipeline of our enterprise accounts.

## 3. Goals and Success Metrics

Primary goals:

- Reduce time-to-insight for enterprise customers from 2 to 3 days to under 10 minutes.
- Achieve 40 percent or higher 30-day retention in the activated pilot cohort by ship.
- Zero critical-severity bugs in the first 30 days post-launch.
- Net Promoter Score of 8 or higher from pilot customers at ship.

Secondary goals:

- Reduce support ticket volume for analytics-related requests by 30 percent.
- Provide a foundation for enterprise API access planned in Q1 2027 (see NS-F011).

## 4. Scope: In-Scope Features

| Feature ID | Feature Name  | Sprint | Status | Pilot Driving          |
|------------|---------------|--------|--------|------------------------|
| NS-F001    | Export to CSV | S12    | Done   | ENT-PC-001, ENT-PC-002 |

| Feature ID | Feature Name                | Sprint | Status      | Pilot Driving |
|------------|-----------------------------|--------|-------------|---------------|
| NS-F002    | Custom date range filter    | S12    | Done        | ENT-PC-003    |
| NS-F005    | User-level drill-down       | S12    | Done        | ENT-PC-002    |
| NS-F003    | Shared dashboard links      | S13    | In progress | ENT-PC-001    |
| NS-F006    | Funnel visualization        | S13    | In progress | All pilots    |
| NS-F004    | Email digest reports        | S14    | Backlog     | ENT-PC-001    |
| NS-F010    | Messaging Alert Integration | S14    | Backlog     | ENT-PC-002    |
| NS-F007    | Cohort comparison table     | S14    | Backlog     | ENT-PC-003    |

Features are cross-referenced in the RICE prioritization workbook (X12.xlsx) and pilot feedback tracker (X07.xlsx). Pilot feedback cycle 4 (October 15, 2026) confirmed that all three accounts endorsed the current in-scope list.

## 5. Feature Specifications

### 5.1 Export to CSV (NS-F001)

Users may export any current dashboard view to a CSV file. The export captures the full visible dataset including applied filters, date range, and cohort selection. Exports are rate-limited to 10 per hour per account to protect backend performance. File size cap: 50 MB. Files are available via a download link that expires in 24 hours. Backend: async job queued in the export-worker service; status polling endpoint provided. Acceptance criteria: export of 100,000 rows completes in under 90 seconds at P95.

### 5.2 Custom Date Range Filter (NS-F002)

Replace the fixed period selector (7 days, 30 days, 90 days) with a calendar picker supporting any range from 1 day to 24 months. The default view remains 30 days. Ranges crossing a fiscal quarter boundary automatically display a quarter-boundary annotation line. Date inputs use ISO 8601 format (YYYY-MM-DD) in the API. Acceptance criteria: custom range persists across page reloads; shared dashboard links preserve the selected range.

### 5.3 User-Level Drill-Down (NS-F005)

From any aggregate metric (e.g., 30-day retention at 41 percent), users may click through to a user-level table showing individual account IDs, event counts, last active date, and cohort membership. PII is masked by default; enterprise admins may unmask for their own tenant only. Pagination: 50 rows per page, max 10,000 rows. Performance requirement: first page loads in under 2 seconds at P95 for cohorts up to 50,000 users.

### 5.4 Shared Dashboard Links (NS-F003)

A signed URL token allows sharing a read-only view of any dashboard configuration with external stakeholders without requiring a Vantage Loop login. Tokens expire in 7 days by default; admins may extend to 30 days. Token scope is limited to the generating account's data. Revocation is instant. Audit log entry created on every token generation and access event.

### 5.5 Funnel Visualization (NS-F006)

Interactive step funnel chart showing conversion between up to 8 user-defined events in sequence. Drop-off percentage displayed at each step. Funnel can be filtered by cohort, date range, and user property. Export to PNG and CSV. Funnel definitions are saved per account and accessible from the dashboard sidebar.

## 6. Out of Scope for This Release

- NS-F008 Annotation layer: deferred to Q1 2027 after pilot cycle 3 showed low demand.
- NS-F009 Benchmark overlay: planned for Q1 2027; requires cross-account data pipeline.
- NS-F011 Enterprise API access: Q1 2027; requires security review and rate-limit design.
- NS-F012 White-label branding: Q2 2027; awaiting brand guidelines from design.
- Mobile-native analytics view: not on roadmap at this time.

## 7. Technical Architecture Overview

Northstar is built as a React frontend consuming a dedicated analytics microservice backed by a ClickHouse analytical database. Event data flows from the main Vantage Loop application through the event pipeline into ClickHouse via a Kafka ingestion pipeline. The analytics service exposes a REST API; all dashboard queries are cacheable at the service layer with a 5-minute TTL. Exports are handled by a separate async worker pool to prevent latency impact on the interactive dashboard.

Authentication uses the identity provider SSO; authorization is role-based within each enterprise account. The shared-link token (NS-F003) uses a signed JWT stored in the analytics service; it does not grant the identity provider session access. All data in transit is TLS 1.3. Data at rest in ClickHouse is encrypted using AES-256. The deployment runs on the container platform (EKS); the monitoring platform monitors latency and error rates; the error-tracking tool captures frontend and backend exceptions.

## 8. Risks and Mitigations

| Risk                                                     | Likelihood | Impact | Mitigation                                                                |
|----------------------------------------------------------|------------|--------|---------------------------------------------------------------------------|
| ClickHouse query latency on large cohorts                | Medium     | High   | Query optimizer review in S13; fallback to sampling for cohorts over 500k |
| Pilot scope creep past feature freeze                    | Medium     | High   | Freeze date locked Oct 9; new requests go to post-launch backlog          |
| Wedding coverage gap Oct 15 to 19                        | Low        | Medium | Megan Shaw holds PM decisions; Carlos Rivera escalation path              |
| the error-tracking tool errors spiking above beta launch | Low        | Medium | Error-rate SLO alert at 0.5 percent; on-call PagerDuty rotation active    |

## 9. Launch Plan Summary

The full launch readiness report is in P09.pdf. Key milestones are:

- October 9, 2026: Feature freeze. All NS-F001 through NS-F010 commits merged.
- October 14, 2026: Beta feedback cycle 4 complete; pilot sign-off obtained.
- October 28, 2026: Release candidate built and smoke-tested.
- November 6, 2026: Load testing complete; the monitoring platform SLOs green.
- November 13, 2026: GA launch. Announcement to all enterprise accounts.
- November 14 onward: 14-day hypercare period; on-call rota doubled.

## 10. Approvals and Sign-Offs

|              |                                                |
|--------------|------------------------------------------------|
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| Data Science | Ryan Iyer                                      |
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